

UF | College of Medicine
UNIVERSITY of FLORIDA



FUSION Demo Session

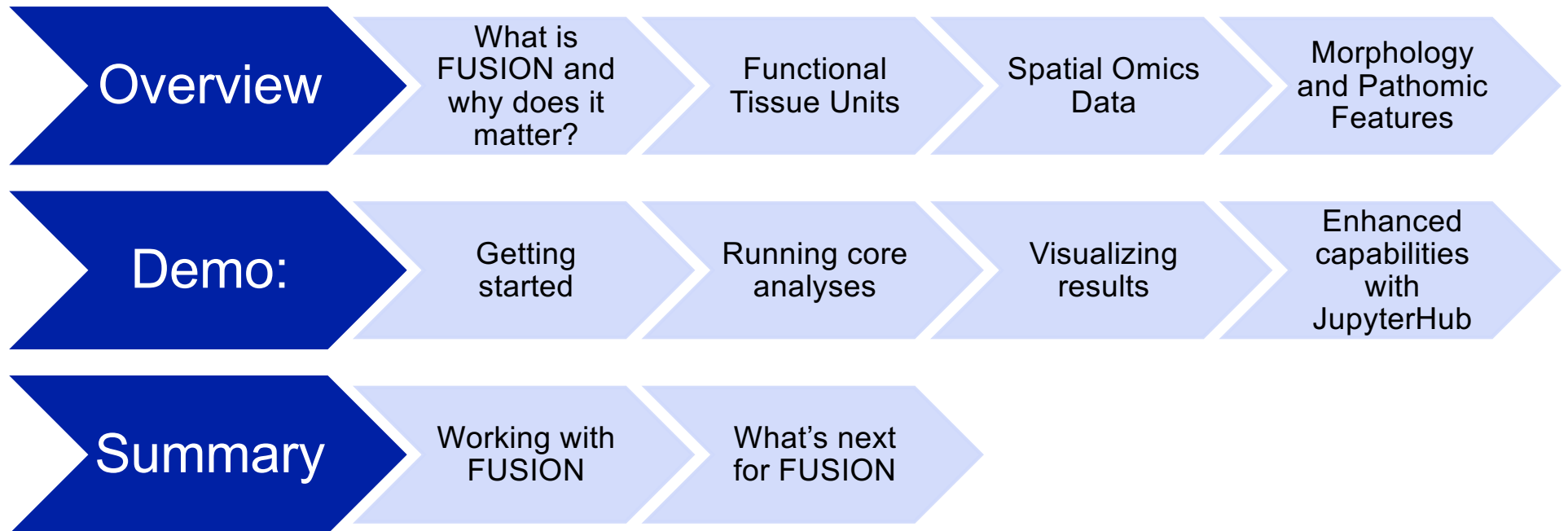
University of Florida

Computational Microscopy Imaging Laboratory

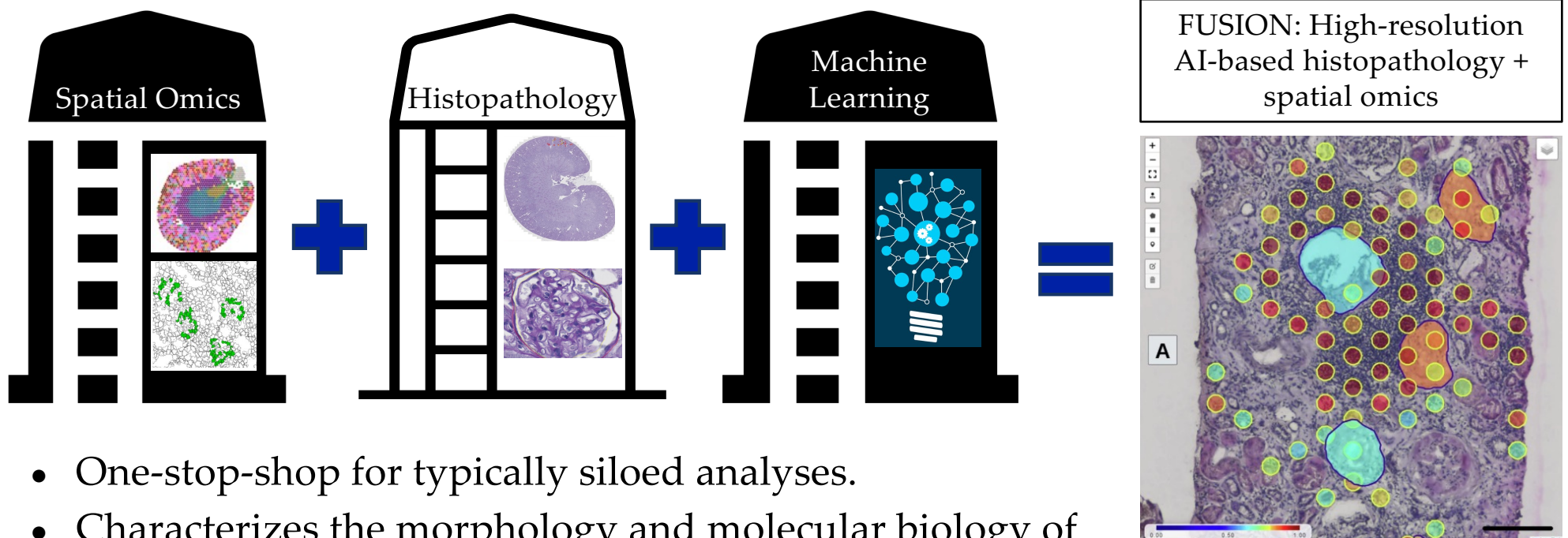
Funding: NIH U54DK134301 and OT2OD033753 (HuBMAP)



FUSION Demo Session Roadmap



What is FUSION: The Core Idea



- One-stop-shop for typically siloed analyses.
- Characterizes the morphology and molecular biology of Functional Tissue Units (glomeruli, tubules, vasculature) in spatial context in high-resolution Whole Slide Image

What do you need to use FUSION?

H&E or PAS-stained whole slide
image (WSI) Frozen or FFPE
tissue



Pre-processed output of a
spatial omics assay run on the
same tissue section (Seurat v5
object in RDS format or Scanpy
AnnData object in h5ad format)

What do you need to use FUSION?

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Pre-processed output of a
spatial omics assay run on
the same tissue section

Multi-Compartment
Segmentation (MCS)



Pathomic Feature
Extraction (PFE)

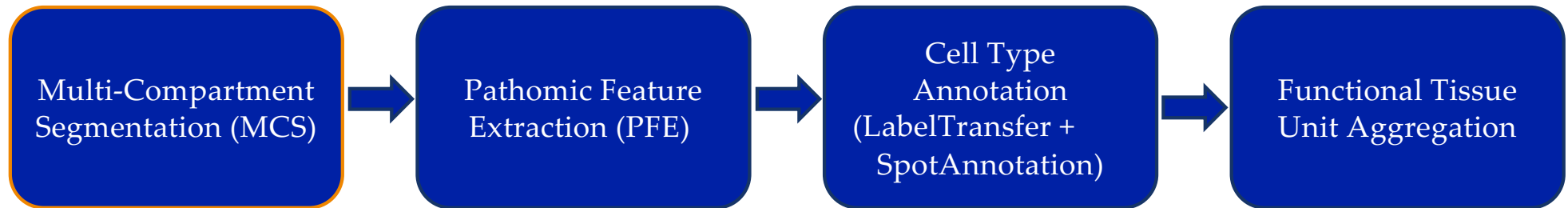


Cell Type
Annotation
(LabelTransfer +
SpotAnnotation)



Functional Tissue
Unit Aggregation

A map of the FUSION analysis pipeline

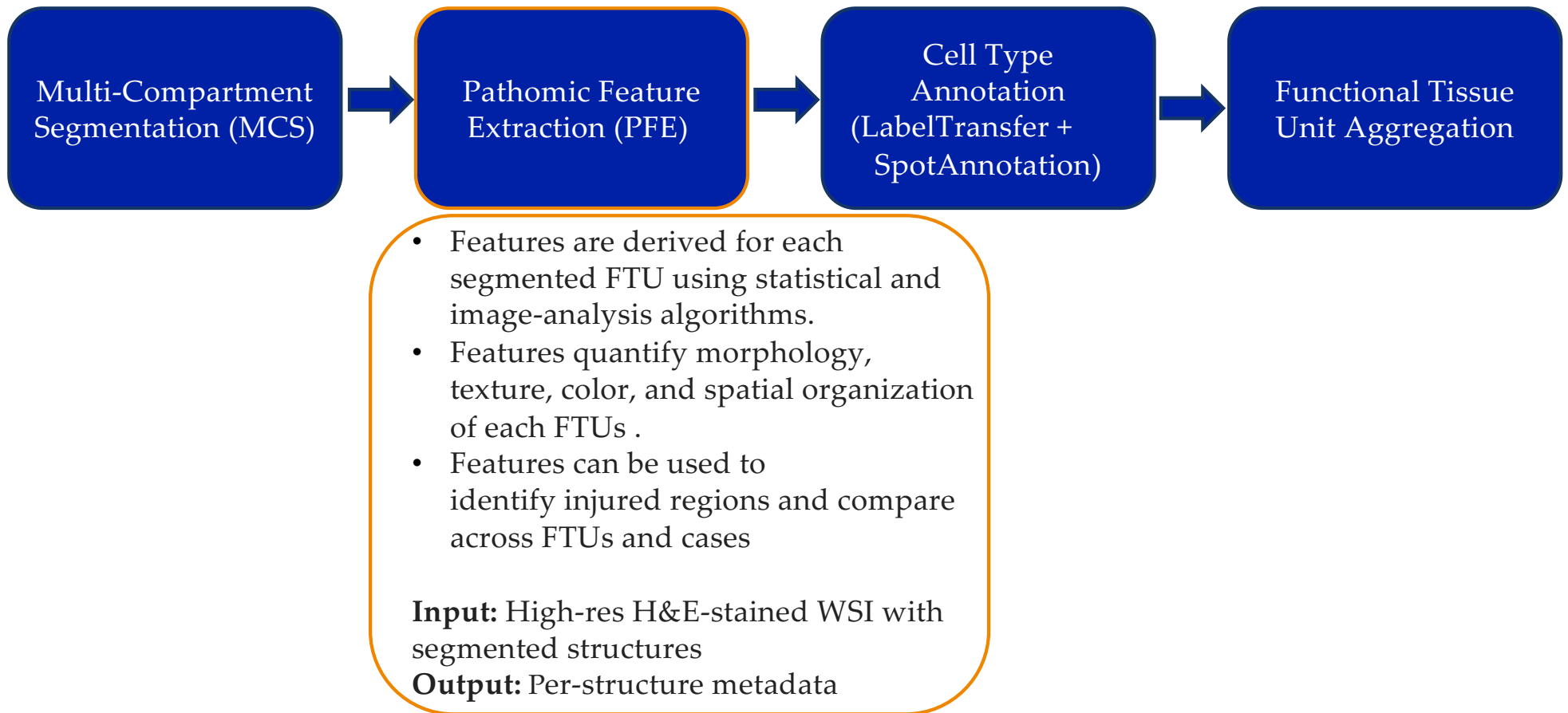


Panoptic deep-learning architecture to perform automated segmentation of six renal compartments, building on AI models trained with high-resolution kidney sections from multiple disease types and stains.

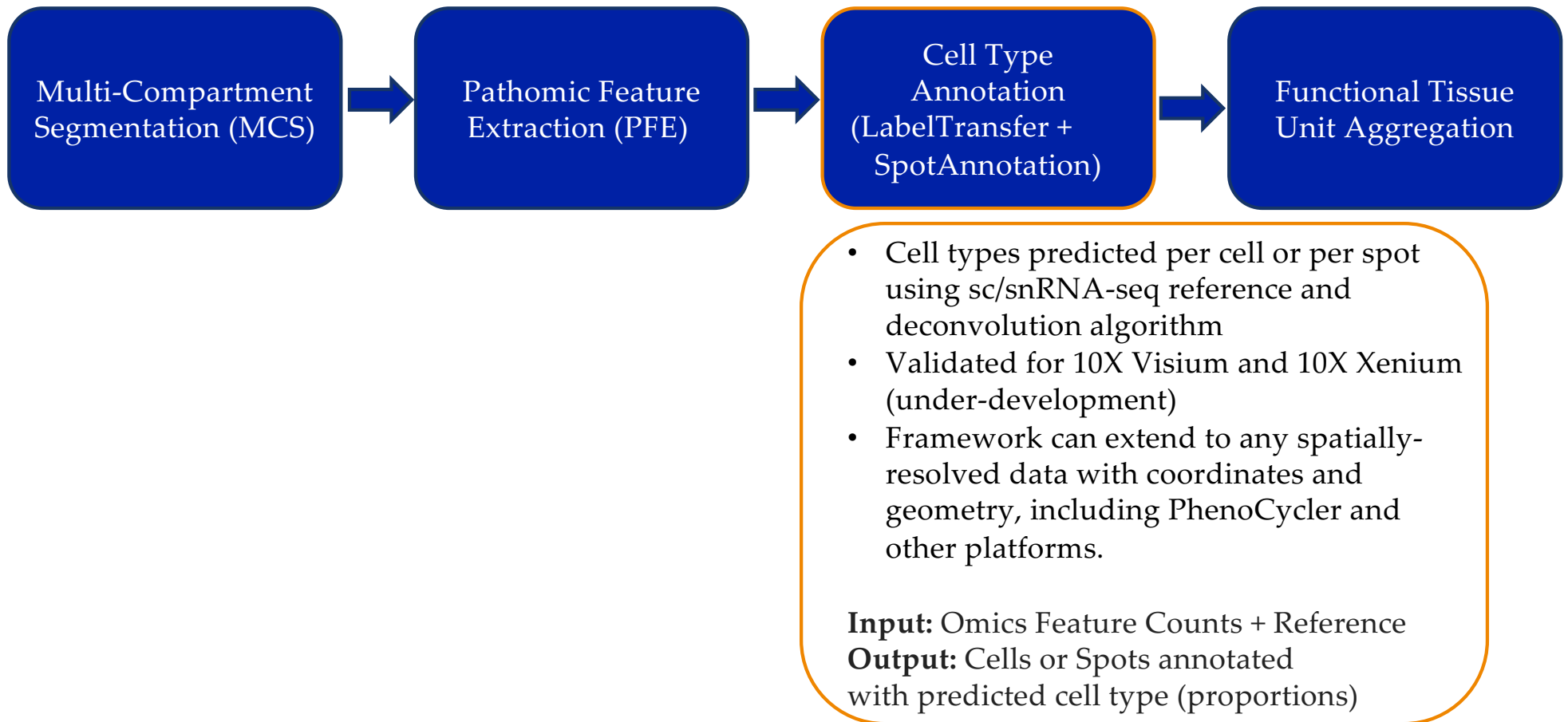
Input: High Resolution H&E or PAS-stained WSI

Output: Segmented structures on WSI: Cortical interstitium, Medullary interstitium, Non-sclerotic glomerulus, Globally sclerotic glomerulus, Tubule, Artery/arteriole

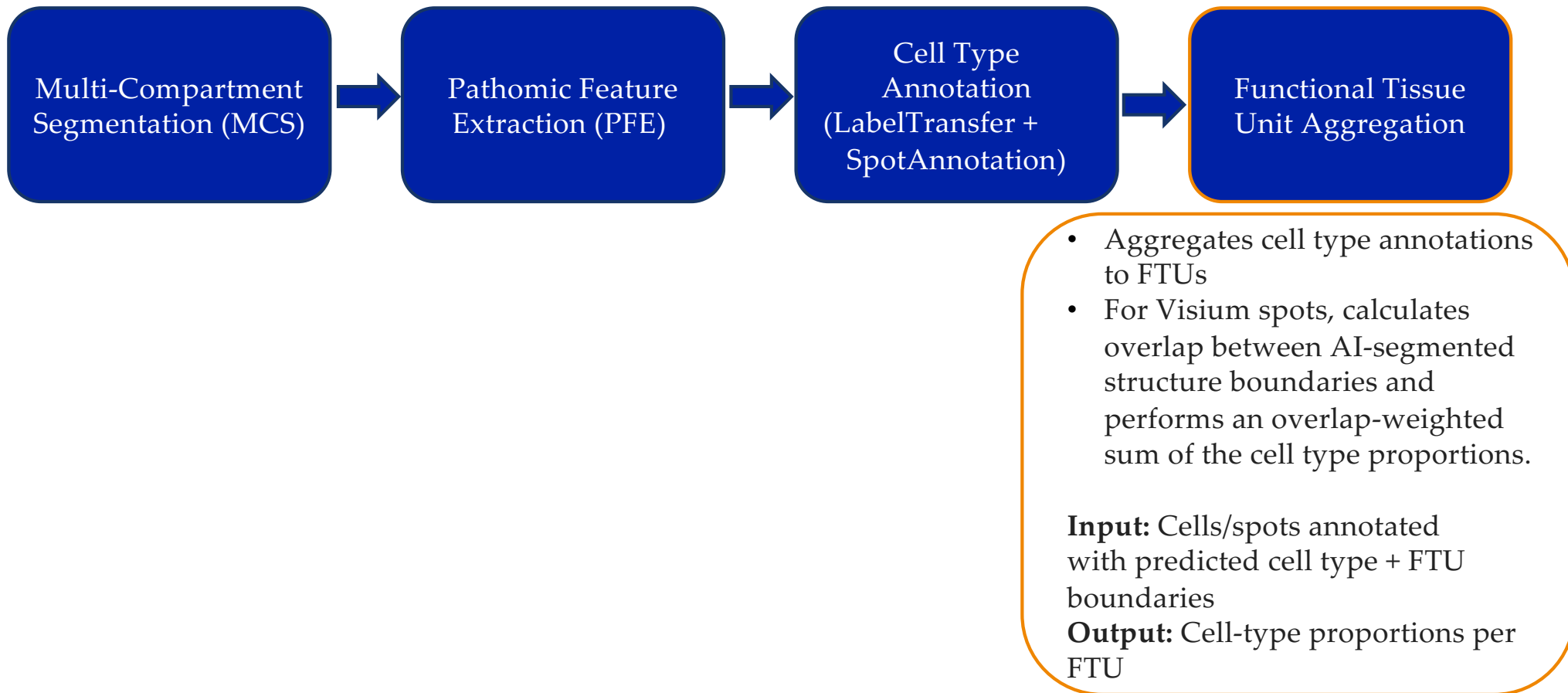
A map of the FUSION analysis pipeline



A map of the FUSION analysis pipeline



A map of the FUSION analysis pipeline



The FUSION Web Application Interfaces

<https://fusionpub.rc.ufl.edu/>

- **FUSION file manager** (powered by the Digital Slide Archive / DSA toolkit):
 - Handles all file storage, organization, and result metadata
- **HistomicsUI**: the interactive visualization layer
 - Accessible via the "Open in HistomicsUI" button top right on any slide.
 - Run analysis plugins, explore results, color FTUs, and generate metadata plots

FUSION Demo: <https://fusionpub.rc.ufl.edu/>

1. [Quantifying glomerulitis in Active Antibody Mediated Rejection and control kidney core needle biopsies \(10X Visium\)](#)
 - Four cases shared by Dr. Yan Chen Wongworawat from Loma Linda University
 - Setup → Demo dataset → Running plugins → Visualizing results → Integration with JupyterHub
2. [Comparing podocyte distribution in two pediatric reference donor biopsies \(1-month-old and 6-year-old\) \(10X Xenium\)](#)
 - Two cases shared by Dr. Sanjay Jain at WashU PCEN
 - Setup → Demo dataset → Running plugins → Visualizing results → Integration with JupyterHub

Getting Started & Data Management

- **[Action: Open Browser and Navigate to <https://fusionpub.rc.ufl.edu/>]** "FUSION is a completely web-accessible platform. You don't need to install local software or manage environments. I am navigating now to the login page. For those of you wanting to explore later, we provide public access to our curated Human Kidney Atlas datasets".
- **[Action: Navigate to 'My Folders' and open the 'PCEN-demo-visium' directory]** "Once inside, you see the FUSION File Manager. This is powered by the **Digital Slide Archive (DSA)** toolkit. It handles all our storage, organization, and result metadata. For this demo we have two slides: Active AMR and control.
- In the AMR folder, you see our primary files: the high-resolution Whole Slide Image, or WSI, which is a .svs file, and several associated data products".
- **[Action: Click green upload button and drag and drop svcs and RDS]** To begin your own analysis, you would click this upload button to upload your WSI and RDS files
- **[Action: Scroll to 'Files & Links' for the upload demo slide]** "Notice that the **Seurat RDS object** is attached directly to the image file here. This file contains your preprocessed omics dataset - the related plugins won't have a data source to pull from. This link ensures that when we later click on a glomerulus, the system knows exactly which Visium spots overlap with that specific geometry".
- **[Action: Click: Open in HistomicsUI]** Once you have your files uploaded, you can open the

HistomicsUI and first plugin, Multi-Compartment Segmentation Plugin

- **[Action: Show key components of sidebar in Histomics UI]** "The right sidebar in histomics UI is where you can review which metadata and annotations you have added, toggle on and off viewing of annotations, and create metadata plots, all of which I will demo".
- **[Action: Click Analyses > sarderlab/fusion_v1> MCS]** "All of the plugins are organized under this analysis button and most run in a few minutes. As I summarized in the introduction, the first step in the analysis pipeline is to run multi-compartment segmentation which performs automated segmentation of six renal compartments, or if you have frozen tissue, this would be frozen glom segmentation".
- **[Action: Scroll to Segmented FTU]** "The output of running this plugin is that these annotations are created in a folder called Segmented FTUs. In this case, they already exist here because I'm not running the plug-in in real time.
- **[Action: Toggle on and off segmented FTU one by one, all]** If you click the eye icon next to an annotation, it will appear on the slide with its default coloring. You can view as many as you want at one time".

Second plugin, Pathomic Feature Extraction

- **[Action: Click Analyses > sarderlab/fusion_v1> PFE]** "This is the 'morphology' power of FUSION. The PFE plugin extracts **72 quantitative features** for every single segmented structure. These features are calculated across three sub-compartments: the **Nuclei**, the **Eosinophilic** (cytoplasmic/matrix) space, and the **Luminal** (void) space.
- **[Action: Expand Metadata]** "After this plugin has been run, its results will be shown here organized by FTU type".
- **[Action: Click on an FTU]** "Practically the most basic way to view these results is by clicking on an FTU and the results will appear in the left hand side bar.
- **[Action: Rectangle select FTUs]** "You can also select multiple FTUs using the rectangle icon and view basic statistical summaries of these same pathomic features for the selected FTUs. Again, there are 72 features so they won't all be immediately familiar or meaningful, but we have supporting documentation to guide their use.
- **[Action: Switch to fusionpub tab, Files & Links]** "These pathomic feature metadata are also saved as excel files so you can open and explore these more carefully. Any of these files can be downloaded by clicking on it"

Third + fourth plugins, Label Transfer + Spot Annotation

- **[Action: Click Analyses > sarderlab/fusion_v1> 10X_VisiumAnalysis > LabelTransfer + Spot Annotation]** "This is where the omics data comes in. We determine a spot's 'molecular identity.' By running **Label Transfer** plugin, which uses **Seurat's TransferData function**. Since a 55-micron Visium spot is a mixture of many cells, we deconvolve those spots using a reference kidney atlas- in this case, the **KPMP Kidney Reference**, but there is an option to allow a user to upload their own reference. This gives us a predicted proportion for every kidney cell type in every spot.
- **[Action: Show Annotations > Visium Spots]** "After this plugin has been run, its results will be shown here with the label Spots...
- **[Action: Click the Visium Spots Annotation]** "And when you toggle them on you can see the Visium spots registered to the WSI.
- **[Action: Click a spot]** "If you click on any on of these spots, you will see its predicted cell type proportions.
- **[Action: Rectangle select multiple spots]** "And if you select multiple spots, just like with FTUs you will see averages of its metadata which for the spots are cell type proportions.
- Now in order to add predicted cell type proportions to the FTUs, we need to run our last plugin so that data from all the cells or spots within the boundaries of an FTU are aggregated.

Fifth plugin: FTU Spot Aggregation

- **[Action: Click Analyses > sarderlab/fusion_v1> 10X_VisiumAnalysis > FTU Spot Aggregation]** "Now in order to add predicted cell type proportions to the FTUs, we need to run our last plug-in so that data from all the cells or spots within the boundaries of an FTU are aggregated."
- **[Action: Show Aggregated FTU folder in sidebar]** After this plug-in runs, FUSION generates a new category of annotations that are aggregated FTUs with one annotation per FTU type.
- **[Action: Toggle on an aggregated FTU]** Now when we click on an FTU, we see its estimated cell type proportions along with its extracted morphometric properties.

Visualize IMM % across glom FTUs with selection and fill+stroke settings

- **[Action: Rectangle select all gloms in case 1 and then case 2]** "Now that we have added all the metadata to our annotations, the simplest way we can get an idea of the molecular and morphological patterns in a slide is to simply explore by selecting structures, in this case glomeruli. Recall that for this demo, we are interested in identifying and scoring glomerulitis in a biopsy from antibody mediated transplant rejection. I know that the region at the top of the slide is from one case, and the bottom is from another. If I select all the gloms in the top case, I can see that there is an average of 50% predicted immune cell infiltration. In the bottom case, this number is much much lower, only 2%.
- **[Action: agg_glom > settings wheel > fill color from IMM blue to red]** To visualize this on a per glomerulus basis, we can color the glomeruli or any FTU based on any of its metadata. Here we will scale the fill from blue to red based on its predicted immune cell proportion so that glomeruli with higher IMM % are more red.

Metadata plots: Violin Plot

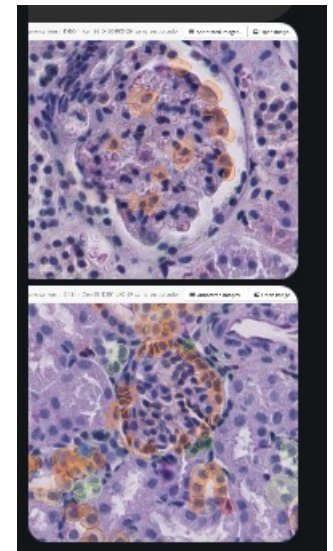
- [Action: Click Gear Icon. Set Format: Violin. Symbol (X): 'Condition'. Y-axis: 'L1 Cell Types - IMM'. CHECK 'Include data from other items in the same folder'. Click Save.] "This is perhaps the most powerful visualization in FUSION 1.0. We are now looking at every single glomerulus from all four slides in our folder—two controls and two AMR cases—all on one graph.
- Look at the results. On the left, the **Control violins** are very narrow and flat; they sit right near the zero line, meaning healthy gloms have almost no immune cell noise. But look at the **Active AMR violins** on the right. They are broadly distributed and significantly 'right-shifted.' This isn't just a simple 'yes' or 'no' diagnosis. FUSION is showing us the **distribution of injury**. We can see that while some gloms in an AMR kidney remain healthy, a distinct sub-population is reaching 50%, 60%, and even 80% immune infiltration. This is a continuous, objective quantification of the **Banff 'g' score**".
- [Action: Select the Lasso tool in the plot and circle the top 3 outlier points in the AMR violin] "FUSION is built for navigation. If I see an outlier point—a glom with extreme inflammation—I don't have to search the slide for it. I simply **lasso those points** in the plot, and the viewer instantly pans and zooms to those exact structures on the WSI. We can see that these three 'molecular outliers' are actually physically clustered in the region of the biopsy with the most severe microvascular injury. This 'Statistical-to-Histological' loop is the core value of

Metadata plots: Scatter Plot

- **[Action: Pathomic Scatterplot: Select Other > non_globally_sclerotic_glomeruli]**
- "We can also use these plots to search for pathomic indicators of disease state. I'm going to look at features related to the size and spread of nuclei, which we would expect to increase in glomerulitis.
- For the X axis I'll select nuclei area and for the y axis Distance Transform Features Sum Distance Transform Nuclei, which essentially measures how spread out the nuclei are within each segmented glom FTU. You can see that both groups follow a tight linear relationship — which makes sense, as larger glomeruli would have both more nuclear area and greater summed distance transform values.
- What's interesting in the context of active AMR is that the circles (AMR) tend to sit above the line formed by the squares (control) at comparable nuclei areas — meaning for a given glomerular size, AMR glomeruli have a higher sum distance transform value.
- This could suggest that in active AMR glomeruli, nuclei are more peripherally distributed or more dispersed within the capillary tuft — which is consistent with the pathology of glomerulitis, where infiltrating leukocytes expand and redistribute within the capillary loops rather than being centrally clustered.
- There are also a few AMR outliers in the upper right with very high values on both axes, suggesting some glomeruli in the AMR cases are substantially enlarged — consistent with our

Xenium

- [Action: Switch tabs to the Xenium Pediatric Donor slide (D451)] "Everything we have seen so far was at the 55-micron Visium spot level. But FUSION is now expanding into **single-molecule resolution with 10x Xenium**. Instead of circles representing mixtures of cells, we now see individual **nuclear and cell boundaries**."
- [Action: Toggle on 'Overview' annotation layers] "In this preview, we are comparing the **spatial distribution of cell types** in two pediatric donors: a 1-month-old (D451) and a 6-year-old (D450)."
- By toggling our 'Overview' layer, we can see the predicted cell types distributed across the WSI.
- The logic of FUSION remains identical: the platform segments the cells, extracts their unique pathomic features, and aggregates the transcript counts within those cell boundaries. Whether the Functional Tissue Unit is a 500-micron glomerulus or a 10-micron macrophage, FUSION allows you to map that data directly back onto the histology."
- [Action: Find a glomerulus in D451 with podocytes peripherally oriented] In the **1-month-old (D451)**, the glomerulus is still relatively immature. The capillary tuft is not yet fully elaborated, so podocytes tend to sit at the **periphery of the glomerular tuft**, lining the outer edge in a more organized, less dispersed arrangement.
- [Action: Find a glomerulus in D450 with podocytes NOT peripherally oriented]
- In the **6-year-old (D450)**, the glomerulus is mature. The capillary network has fully branched and expanded, and podocytes have spread out to follow those loops throughout the tuft — so they appear **more distributed and less peripherally concentrated**, which in a 2D section can make them look scattered or even central.



FUSION++: FUSION as an API with Enhanced Capability

Access FUSION from HuBMAP workspace.

Demo Agenda

1. From FUSION to FUSION++: *The Power of Integration*
2. Challenges of FUSION++: *Navigating the Complexity*
3. How to access HuBMAP workspace
4. The quick start
5. Analysis in Action
6. Visual Analytics: *Interactive Annotation & Plotting*
7. Beyond the Basics: *Expanding the Horizon*

From FUSION to FUSION++: *The Power of Integration*

- **Reproducible workflow:** Fetch data, run analysis, visualize, annotate, and plot.
- **Collaboration:** Bring together biologists, bioinformaticians and data scientists.
- **HuBMAP Compute:** Leverages the massively scalable HuBMAP infrastructure.
- **Secure Enclave:** User data remains private.
- **Training Platform:** Great teaching resource.

Challenges of Fusion++: *Navigating the Complexity*

Code-Centric Interface

Starter demo-notebooks
to support initial
adoption and usage

How to access HuBMAP Workspace

Prerequisite: HuBMAP workspace access – <https://hubmapconsortium.org/workspaces-sign-up-9362/>

The screenshot shows the HuBMAP workspace management interface. At the top, there's a dark blue header with the HuBMAP logo, 'Data' and 'Resources' links, and a user profile icon. Below the header, a notification bar states: 'Manage your workspace collaboration with notifications for invitations shared with you or sent by you. For received invitations, you can preview the workspace details before deciding to accept or decline the invitation.' Below this, there are tabs for 'Received' and 'Sent'. The 'Received' tab is active, showing a message: 'No received workspace invitations.' Below this, the section 'My Workspaces (2)' is displayed. It includes a search bar 'Search workspace by name or ID', a trash icon, a share icon, a 'Stop Workspaces' dropdown, and a '+ Create New' button (highlighted with a red box). Below the search bar, there are filters: '0 selected', 'Created by Me (2)', and 'Created by Other (0)'. A table lists the workspaces:

☐	Name ↓	Status ↓	Created By ↓	Last Launched ↓	
☐	new-WSI viz (ID: 1815)	active ✓	Me	2026-01-19	Stop Launch
☐	Jupyter-hub (ID: 1813)	idle ⓘ	Me	2025-12-18	Launch

 Advanced configuration settings are not retained from previous sessions. If you customized settings in a previous launch, you will need to reapply those changes.

Environment Selection

All workspaces environments have built-in Python support, with the option to include R or pre-installed packages as needed. To view the full list of packages, use the `help('modules')` command for Python or `install.packages()` for R in your workspace.

If a workspace was previously launched with R, launching a workspace without R support may cause issues with your saved work.

Please note that selecting environments with extra features may result in longer loading times. For environments with GPU packages, it is strongly recommended to enable GPU support in the advanced configurations section. Since GPU resources are limited, please select this option only if it is essential for your workspace.

If you have questions about these environments and their respective packages or suggestions for improvements, please [contact us](#) .

Select Environment

- ☐ JupyterLab (Python - Base) 
- ☐ JupyterLab (Python+R) 
- ☐ JupyterLab (Python - Non-GPU Common Packages) 
- ☒ JupyterLab (Python - GPU Common Packages) 
- ☐ JupyterLab (Spatial Data Workshop)

Advanced Configurations (Optional)

Cancel

Launch Workspace

What's next for FUSION (2.0)

- Integration of other spatial omics and multimodal assays
- Support for multiple cell type annotation strategies
- Plugin based differential expression and niche analysis



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